

Manyata Singh

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Bhilai / Durg, Chhattisgarh, India

Professional Summary

Pre-final year B.Tech CSE student specializing in Generative AI, LLM engineering, and agentic workflow development. Designed and shipped production-ready AI applications including a hierarchical multi-agent finance chatbot and a RAG-powered career advisor — both deployed end-to-end with real API integrations. Holds 5 Google Cloud GenAI certifications. Actively exploring LoRA/QLoRA fine-tuning, vector databases, and voice AI. Seeking an AI/ML internship to contribute meaningfully to a product engineering team from day one.

Technical Skills

- **Languages & Frameworks:** Python, JavaScript, HTML/CSS, React.js, Flask, Streamlit, Node.js, Express.js
- **AI / ML:** LLM Prompt Engineering, RAG Pipelines, LangChain, LlamaIndex, Agentic Workflows, NLP (TF-IDF, Embeddings), Model Fine-tuning (LoRA/QLoRA basics), Multi-turn Dialogue Design
- **LLM Platforms:** OpenAI GPT-4, Google Gemini, Anthropic Claude, HuggingFace Transformers
- **Cloud & Infra:** Google Cloud Platform (GCP), Firebase, Docker (basics), REST APIs, Git/GitHub, Postman, Cloud Deployment (Streamlit Cloud)
- **Databases & Tools:** PostgreSQL (learning), MongoDB, Vector DBs (Pinecone, ChromaDB), Cloudinary, Jupyter Notebook, VS Code, Notion
- **Concepts:** Retrieval-Augmented Generation, Prompt Versioning, A/B Testing, OCR, Voice AI (LiveKit, Whisper), LLMops (Langfuse)

Experience

Independent AI/ML Engineer

April 2025 – Present

Self-Directed Projects & Research

Remote

- Engineered and deployed 3+ AI-powered applications using Python, LangChain, Streamlit, and OpenAI APIs, covering the full development lifecycle from architecture to cloud deployment.
- Designed multi-turn conversational prompt flows with goal-oriented dialogue management for voice and chat interfaces, improving response coherence across long sessions.
- Implemented RAG pipelines that integrated domain-specific knowledge bases, measurably reducing LLM hallucinations in specialized advisory tasks.
- Applied prompt engineering techniques (chain-of-thought, few-shot, role prompting) and A/B tested prompt variants to optimize output quality and latency.

Product & Content Contributor

2024

PeerX – Student Opportunity Aggregator (Shark Tank India S5)

Remote

- Contributed to entrepreneurial initiative feature development and product ideation at an early-stage startup recognized on Shark Tank India Season 5.
- Managed social media strategy and content pipeline, growing platform visibility and driving user engagement for a student-focused opportunity aggregator.

Projects

Agentic Finance Chatbot · *Python, Flask, LangChain, OpenAI API*

- Architected a hierarchical multi-agent system comprising a Research Agent (data fetching), Analysis Agent (processing), and Editor Agent (summary generation) — mirroring production-grade agentic finance workflows.
- Implemented multi-turn conversation with persistent context window management to maintain full dialogue state across sessions, reducing context loss errors.
- Engineered voice-optimised prompts that explicitly spell out numbers, dates, and IDs for TTS-safe output; used LangChain for agent orchestration, tool-use routing, and prompt versioning across sub-agents.

AI Career Advisor · *Python, Streamlit, TF-IDF, RAG, OpenAI API*

- Built an AI-powered career roadmap advisor that generates personalised 12-week learning plans by combining TF-IDF cosine similarity retrieval with OpenAI API calls; deployed live on Streamlit Cloud.
- Integrated chain-of-thought and few-shot prompt engineering, improving output relevance and specificity for domain-targeted career advice.
- Added a RAG component that dynamically retrieves career resources to ground LLM responses, reducing hallucinations in specialized advisory contexts; optimised API latency via caching and batched requests.

CloudVault · *React.js, Node.js, Express.js, MongoDB, Firebase, Cloudinary*

- Built a full-stack cloud storage platform with Firebase Authentication, drag-and-drop file uploads, real-time upload progress tracking, Cloudinary cloud storage integration, and features including file preview, search, and category filtering.
- Engineered a Secret Vault module with AES-256 encryption, SHA-256 PIN hashing, and brute-force lockout protection; built a browser-side image compression tool with live before/after size comparison for bandwidth optimisation.
- Designed a fully responsive UI with dark/light mode, glassmorphism styling, and toast notifications; secured the REST API backend with rate limiting, Helmet security headers, and MVC architecture — with roadmap for folder organisation, shareable file links, OAuth login, and React Native mobile app.

Education

Bhilai Institute of Technology, Durg

2023 – 2027

B.Tech – Computer Science & Engineering | CPI: 8.45 / 10.0

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Cloud Computing, Web Development.
- Selected for SIH 2025 internal hackathon; participant in multiple external hackathons with experience presenting AI solutions to technical and non-technical audiences.

Techonet, Durg, Chhattisgarh

August 2024

Vocational Training – Artificial Intelligence

- Completed hands-on training in data preprocessing, supervised & unsupervised ML, NLP basics, model evaluation, and deployment pipelines on real-world datasets.

Certifications

- **Develop GenAI Apps with Gemini and Streamlit** – Google Cloud Skills Boost *May 2024*
- **Prompt Design in Vertex AI** – Google Cloud Skills Boost *May 2024*
- **Machine Learning Operations (MLOps) for Generative AI** – Google Cloud Skills Boost *April 2026*
- **Machine Learning Operations with Vertex AI: Model Evaluation** – Google Cloud Skills Boost *April 2026*
- **Responsible AI for Developers: Fairness & Bias** – Google Cloud Skills Boost *April 2026*

Additional

- **Currently Exploring:** Fine-tuning open-source LLMs (LoRA/QLoRA), vector databases (Pinecone, ChromaDB), voice AI (LiveKit, Whisper), and LLMOps (Langfuse tracing).
- **Interests:** Generative AI, Agentic Systems, Startup Building, Astronomy & Space Science, Chess (strategic thinking).